

DATABASE SYSTEMS

1. Entity-Relationship (ER) Model

- Concepts of Entities, Attributes, and Relationships
- Cardinality Constraints
- Weak Entities and Identifying Relationships
- ER Diagram Representation

2. Enhanced Entity-Relationship (EER) Model

- Generalization and Specialization
- Aggregation
- Superclasses and Subclasses
- EER Diagram Representation

3. Relational Model

- Relation Schema and Instances
- Keys (Primary Key, Foreign Key)
- Domain and Tuple
- Relational Integrity Constraints
- Relational Operations (Select, Project, Join, etc.)

4. SQL Operations

- **Create**: Creating tables, views, indexes
- **Insert**: Inserting data into tables
- **Delete**: Deleting records from tables
- **Update/Alter**: Modifying existing records, schema changes (ALTER TABLE)
- **Aggregate Functions**:
 - SUM, AVG, COUNT, MIN, MAX
- **Clauses**:
 - WHERE: Filtering records
 - HAVING: Filtering aggregated records
 - ORDER BY: Sorting results

5. View

- Concept of Views
- Creating Views
- Using Views in Queries
- Modifying Data Through Views

6. Normalization

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)
- Decomposition and Dependency Preservation

7. Generalization

- Concept and Implementation
- Use in the context of EER Models
- Hierarchical Structure of Generalization